Basic Mathematics - VDPP

Questions

1. If $a - \frac{1}{a} = 6$, then find $a^2 + \frac{1}{a^2}$.

(a) 36

(b) 38

(c) 40

(d) 42

2. If $a - \frac{1}{a} = 6\ find\ a^4 + \frac{1}{a^4}$

(a) 1442

(b) 1443

(c) 1444

(d) 1445

3. If $a + \frac{1}{a} = \sqrt{2}$; Find $a^2 + \frac{1}{a^4}$.

(a) 0

(b) –1

(c) 2

(d) no solution

4. If $x^2 + \frac{1}{x^2} = 83$, Find the value of $x - \frac{1}{x}$

(a) ± 8

(b) 8

(c) ± 9

(d) 9

5. If $x^2 + \frac{1}{x^2} = 34$,then find the value of $x + \frac{1}{x}$

(a) ± 6

(b) ± 3

(c) ± 5

(d) ± 4

6. Solve $\left(x^2 + \frac{1}{x^2}\right) - 35\left(x + \frac{1}{x}\right) + 62 = 0.$

(a) $2, \frac{1}{2}, 3, \frac{1}{3}$

(b) $1, 2, 3, \frac{1}{3}$

(c) $2, \frac{1}{2}, 3, 4$

(d) $2, \frac{1}{2}, \frac{1}{3}, 4$

7. Solve $x^2 + \frac{1}{x^2} - 3x - \frac{3}{x} + 4 = 0.$

(a) $x = \pm 1$

(b) x = -1, 2

(c) x = 1

(d) x = 1, -2

8. Solve $2x^2 - 9x + 14 - \frac{9}{x} + \frac{2}{x^2} = 0$

(a) 2, 3, 1

(b) 2, 3, $\frac{1}{2}$

(c) 2, $\frac{1}{2}$, 1

(d) $\frac{1}{3}$, 2, 1

9. Solve $3x^4 - 4x^3 - 14x^2 - 4x + 3 = 0$

(a) $3, \frac{1}{3}, 1$

(b) $3, \frac{1}{3}, -1$

(c) $3, \frac{1}{2}, 1$

(d) $3, \frac{1}{2}, -1$

10. Solve $x^2 + \frac{1}{x^2} - 3\left(x - \frac{1}{x}\right) = 0.$

(a) $\frac{1 \pm \sqrt{5}}{2}, 1 \pm \sqrt{2}$

(b) $\frac{1 \pm \sqrt{4}}{2}, 1 \pm \sqrt{2}$

(c) $\frac{1 \pm \sqrt{5}}{2}, 1 \pm \sqrt{3}$

(d) $\frac{1 \pm \sqrt{4}}{2}, 1 \pm \sqrt{3}$

11. Solve $x^4 - 4x^3 + x^2 + 4x + 1 = 0.$

(a) $\frac{1 \pm \sqrt{3}}{2}, \frac{3 \pm \sqrt{4}}{2}$

(b) $\frac{1 \pm \sqrt{5}}{2}, \frac{3 \pm \sqrt{3}}{2}$

(c) $\frac{1 \pm \sqrt{5}}{2}, \frac{3 \pm \sqrt{13}}{2}$

(d) $\frac{1 \pm \sqrt{5}}{2}, \frac{3 \pm \sqrt{4}}{2}$

12. If $x^2 + a^2 + 9 - xa - 3a - 3x = 0$ then

(a) x = a or a = 1

(b) x = a and a = 3

(c) x = ± and a = 3

(d) none of these

13. If $4a^2 + 9b^2 + 16c^2 = 6ab + 8ac + 12bc$, then _____.

(a) $2a = 3b = 4c$

(b) $2a = 3b \neq 4c$

(c) $2b \neq 3b = 4c$

(d) $2a \neq 3b \neq 4c$

14. If $4x^2 + 9y^2 + z^2 = 6xy + 3yz + 2xz$then

(a) x = 2y = 3z

(b) 2x = 3y = z

(c) x = y = z

(d) none of these

15. If $a^2 + \frac{b^2}{4} + 4c^2 - \frac{ab}{2} - bc - 2ac = 0$,then _____.

(a) a = b = 2c

(b) a = b = c

(c) $a = \frac{b}{2} = 2c$

(d) $a = \frac{b}{2} = c$

16. If $a^2 + b^2 + c^2 = abc\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right)$then

(a) a = b = c

(b) a = 2b = 3c

(c) 2a = b = c

(d) None of these

17. If $a + \frac{1}{a} = 4$ then find $a^2 + \frac{1}{a^2}$.

(a) 16

(b) 18

(c) 14

(d) 12

18. If $a + b + c = 6$ and $a^2 + b^2 + c^2 = 24$. Find $ab + bc + ca$.

(a) 12

(b) 6

(c) 8

(d) 10

19. If $x^2 + 4y^2 + 16 = 2xy + 4x + 8y$ then

(a) x = 4 & y = 4

(b) x = 2 & y = 4

(c) x = 4 & y = 2

(d) x = 2 & y = 2

20. If $a+2b+c=8$ & $a^2+4b^2+c^2=50$ find $2ab+2bc+ac.$

(a) 9

(b) 7

(c) 14

(d) 6

21. If $(a-b)^2+(b-c)^2+(c-a)^2=0$, then

(a) a + b + c = 0

(b) a = b = c

(c) $ab+bc+ca=0$

(d) None of these

22. If a > b, then

(a) a + 5 > b + 5

(b) a – 5 < b - 5

(c) a + b < b + b

(d) Depends on a & b

23. If a > b, then :

(a) 2a > 2b

(b) a – 5 < b - 5

(c) -5a > - 5b

(d) None of these

24. If a < b, then:

(a) $\frac{a}{(-2)}<\frac{b}{(-2)}$

(b) $\frac{a}{2}>\frac{b}{2}$

(c) $\frac{1}{a}<\frac{1}{b}$

(d) $\frac{a}{-2}>\frac{b}{-2}$

25. If $\log_{\sqrt{8}}b=3\frac{1}{3}$, then b is equal to _____

26. Evaluate $3\log_{10}1.5+\log_{10}240-2\log_{10}9$

(a) 1

(b) 0

(c) $\ln 10$

(d) None of these

27. Evaluate $3\log 4-2\log 6+\log(18)^{3/2}$

(a) 0

(b) $\log(96\sqrt{2})$

(c) 1

(d) $\log(192\sqrt{2})$

28. Solve for x : log 3 + log (x + 1) = log (2x + 7)

(a) 1

(b) 3

(c) 4

(d) 6

29. Solve $\log 2 + \log(x+2) - \log(3x-5) = \log 3.$

(a) $x = \frac{19}{7}$

(b) x = 7

(c) $x = \frac{7}{19}$

(d) x = 19

30. Solve $\log_4 \log_3 \log_2 x = 0$

(a) 2

(b) 3

(c) 6

(d) 8

31. The value of $\log_{10} 1000 - \log_{10} 100$ is

(a) 0

(b) 1

(c) 13

(d) 2

32. Which of the following is equal to $\log_{10}5 + \log_{10}20$?

(a) $\log_{10}25$

(b) 2.5

(c) 2

(d) none of these

33. $\frac{\log_2 7}{1+\log_2 3}$ = ?

(a) $\frac{\log_2 6}{\log_2 7}$

(b) $\frac{\log_2 7}{\log_2 6}$

(c) $\log_2 7 \times \log_2 6$

(d) none of these

34. If $2\log_4 3 + \log_4 \frac{x^2}{9} = 1$, then the value of |x| is equal to _____.

(a) 4

(b) 2

(c) 9

(d) none of these

35. Solve for x : 2 + log x = log 500

(a) 5

(b) 100

(c) 500

(d) none of these

36. Logarithm of $32\sqrt[5]{4}$to the base $2\sqrt{2}$ is

(a) 3.6

(b) 5

(c) 5.6

(d) None of these

37. The value of $\log_3 4 \cdot \log_4 5 \cdot \log_5 6 \cdot \log_6 7 \cdot \log_7 8 \cdot \log_8 9$ is

(a) 1

(b) 2

(c) 3

(d) 4

38. $\log_7 \log_7 \sqrt{7\sqrt{7\sqrt{7}}} =$

(a) 3 $\log_2$ 7

(b) 1 – 3$\log_3$7

(c) 1 – 3 $\log_7$2

(d) None of these

39. If $a^2 + 4b^2 = 12ab$, then log (a + 2b) is

(a) $\frac{1}{2}[\log a + \log b - \log 2]$

(b) $\log \frac{a}{2} + \log \frac{b}{2} + \log 2$

(c) $\frac{1}{2}[\log a + \log b + 4\log 2]$

(d) $\frac{1}{2}[\log a - \log b + 4\log 2]$

40. If A = $\log_2 \log_2 \log_4 256 + 2\log_{\sqrt{2}} 2$, then A is equal to

(a) 2

(b) 3

(c) 5

(d) 7

41. If $\frac{\log x}{b-c}=\frac{\log y}{c-a}=\frac{\log z}{a-b}$, then which of the following is incorrect?

(a) xyz = 0

(b) $x^a y^b z^c = 1$

(c) $x^{b+c} y^{c+a} z^{a+b} = 1$

(d) $xyz = x^a y^b z^c$

42. The solution of the equation $\log_7 \log_5 \left(\sqrt{x^2+5+x}\right)$= 0 is

(a) x = 2

(b) x = 3

(c) x = 4

(d) x = -2

43. If $3^{2\log_3 x} - 2x - 3 = 0$, then the number of values of 'x' satisfying the equation is

(a) 0

(b) 1

(c) 2

(d) more than 2

44. Find the value of x : $\log_{10}\left(x^2-12x+36\right)=2$

(a) 12

(b) 16

(c) 14

(d) 11

45. Find the value of x : $\log_3\left(\log_9 x+\frac{1}{2}+9^x\right)=2x$

(a) $\frac{1}{3}$

(b) $\frac{2}{3}$

(c) 3

(d) 1

46. If $x^2 > 4$, then

(a) x > 2

(b) $x \in (-\infty,-2)\cup(2,\infty)$

(c) -2 < x < 2

(d) None of these

47. Solve for x :

(x – 1) (x – 3) > 0

(a) $x \in (-\infty, -2) \cup (2, \infty)$

(b) $x \in (-\infty, 1] \cup (3, \infty)$

(c) $x \in (-\infty, -2) \cup (2, \infty]$

(d) $x \in (-\infty, 1) \cup (3, \infty)$

48. Solve for x :

(x − 8) (x − 7) < 0

(a) x ∈ (7, 5)

(b) x ∈ (7, 8)

(c) x ∈ (6, 7)

(d) x ∈ (7, 9)

49. Solve for x:

x^2 − 5x + 6 ≥ 0

(a) $x \in (-\infty, 2) \cup [3, \infty)$

(b) $x \in (-\infty, 2) \cup (3, \infty)$

(c) $x \in (-\infty, 2] \cup (3, \infty)$

(d) $x \in (-\infty, 2] \cup [3, \infty)$

50. Solve for x :

x^2 − 1 ≤ 0

(a) $x \in (-1, 1)$

(b) $x \in [-1, 1]$

(c) $x \in [-1, 1)$

(d) $x \in (-1, 1]$

51. solve for x :

(x − 3) ≥ (x − 3) (x − 4)

(a) $x \in [3, 6]$

(b) $x \in [3, 5]$

(c) $x \in (3, 6)$

(d) $x \in (3, 5)$

52. Solve for x :

$\frac{x+5}{x} \geqslant 1$

(a) $x \in (-1, \infty)$

(b) $x \in (-3, \infty)$

(c) $x \in (-2, \infty)$

(d) $x \in (0, \infty)$

53. Solve

$\frac{2}{x-3} \leqslant 5$

(a) $x \in (-\infty, 3) \cup \left(\frac{17}{5}, \infty\right)$

(b) $x \in (-\infty, 3] \cup \left[\frac{17}{5}, \infty\right)$

(c) $x \in (-\infty, 3) \cup \left[\frac{17}{5}, \infty\right)$

(d) $x \in (-\infty, 3] \cup \left(\frac{17}{5}, \infty\right)$

54. Solve for x :

$x^2 + 4x + 4 \leq 0$

(a) x = - 2

(b) x = 0

(c) $x \in \varphi$

(d) $x \in R$

55. Solve for x :

$x^2 + 4x + 4 > 0$

(a) R – {2}

(b) x = -2

(c) x =2

(d) R – {-2}

56. Solve for x :

$(x-1)(x-4)(x-6)(x-9) \leqslant 0$

(a) $x \in [1, 4] \cup [6, 9]$

(b) $x \in (1, 4) \cup [6, 9]$

(c) $x \in [1, 4] \cup (6, 9)$

(d) $x \in [1, 4) \cup [6, 9]$

57. Solve for x :

(2x – 1) (x – 1) (x + 8) (x – 7) ≤ 0

(a) $x \in \left[-8, \frac{1}{2}\right] \cup [1, 7]$

(b) $x \in [-8, 2] \cup [1, 7]$

(c) $x \in \left[-8, \frac{1}{2}\right] \cup \left[1, \frac{1}{7}\right]$

(d) $x \in \left[-\frac{1}{8}, \frac{1}{2}\right] \cup [1, 7]$

58. Solve for x:

$\frac{(x+4)(x-1)(x+5)}{x(x-7)} \leqslant 0$

(a) $x \in (-\infty, -4] \cup [-4, 0) \cup [1, 7)$

(b) $x \in (-\infty, -5] \cup [-4, 0) \cup [1, 7)$

(c) $x \in (-\infty, -5] \cup [-4, 1) \cup [1, 7)$

(d) $x \in (-\infty, -5] \cup [-4, 2) \cup [1, 7)$

59. Solve for x:

$$\frac{x^2 - 5}{x - 1} \geqslant -1$$

(a) $x \in [-3, -1) \cup [2, \infty)$

(b) $x \in [-3, 1) \cup [-2, \infty)$

(c) $x \in [-3, 1] \cup [2, \infty)$

(d) $x \in [-3, 1) \cup [2, \infty)$

60. $x^2 (x - 5) (x - 2) \geq 0$

(a) $x \in (-\infty, 2] \cup [5, \infty)$

(b) $x \in (-\infty, 2) \cup [5, \infty)$

(c) $x \in (-\infty, -2] \cup (5, \infty)$

(d) $x \in (-\infty, 2) \cup (5, \infty)$

61. $\frac{x^2 (x - 5)^4}{(x - 2)} < 0$

(a) $x \in (-\infty, 2) - \{0\}$

(b) $x \in (-\infty, 2] - \{0\}$

(c) $x \in [-\infty, 2)$

(d) $x \in [-\infty, 2]$

62. $(x - 2)^3 (x - 4)^2 (x - 10)^{17} \geqslant 0$

(a) $x \in (-\infty, 2] \cup (10, \infty)$

(b) $x \in (-\infty, 2) \cup [10, \infty)$

(c) $x \in [-\infty, 2] \cup [10, \infty)$

(d) None of these

63. $(x - 2)^4 (x - 3)^2 (2x + 1)^{24} \leq 0$

(a) $x = 2, 3, \frac{-1}{2}$

(b) $x = 2, 3, \frac{1}{2}$

(c) $x = 1, 3, \frac{-1}{2}$

(d) $x = 1, 2, \frac{-1}{2}$

64. $\frac{(2x + 1)^3 (x + 4)^2}{(x - 5)^5 (x - 17)^2} \geqslant 0$

(a) $x \in \left(-\infty, \frac{-1}{2}\right) \cup (5, 17) \cup (17, \infty)$

(b) $x \in \left(-\infty, \frac{-1}{2}\right] \cup [5, 17] \cup (17, \infty)$

(c) $x \in \left(-\infty, \frac{-1}{2}\right] \cup (5, 17) \cup (17, \infty)$

(d) $x \in \left(-\infty, \frac{-1}{2}\right] \cup (5, 17) \cup [17, \infty)$

65. $\left(x^2+1\right)(x-2)^2(x-3) < 0$,then x belongs to:

(a) $(-\infty, 2) \cup (2, 3)$

(b) $(-\infty, 3)$

(c) (2, 3)

(d) none of these

66. Which of the following expressions is equal to $\left|x^2+1\right|$?

(a) $-(x^2+1)$

(b) x^2+1

(c) $-x^2+1$

(d) None of these

67. Let A denotes the set of values of x for which $\frac{x+2}{x-4} \leqslant 0$ and B denotes the set of values of x for which $x^2 - ax - 4 \leqslant 0$. If B is the subset of A, then find the number of possible integral values of a.

68. If $\left|x^2-12x+32\right| + \left|x^2-9x+20\right| = 0$. Then find the value of x.

69. What is the minimum values of the function $|x-4|+2$?

70. Solve for x : |x − 3| = 5

(a) x = 8, − 2

(b) x = 8, 2

(c) x = 7, − 2

(d) x = 7, 2

71. Solve for x : |2x − 3| = −5

(a) x = 1

(b) x = 2

(c) x = 3

(d) No Solution

72. Solve for x : $|x^2 + 1| = 5$

(a) x = ± 3

(b) x = − 2

(c) x = ± 2

(d) x = − 3

73. If |x − 2| = 3 then values of x are

(a) 3 or − 3

(b) 5 or − 1

(c) − 5 or 1

(d) − 4 or 2

74. If |x + 5| = 6, then values of x are

(a) 1 or − 11

(b) − 1 or 11

(c) − 6 or 6

(d) 3 or − 2

75. If $|x^2 - 4x + 2| = 2$ then values of x are

(a) 1 or 3 or 5

(b) 0 or 2 or 4

(c) 5 or 7

(d) 12 or 14

76. If $|x^2 - 1| = 1$, then values of x are

(a) ± $\sqrt{2}$ or 0

(b) ± $\sqrt{3}$ or 0

(c) ± 2 or 1

(d) None of these

77. The values of x satisfying |x + 3| = −3 is/are

(a) 6 or 0

(b) 3 or − 3

(c) 0 or − 6

(d) None of these

78. If |x| = 3 then x can be

(a) 2

(b) 3

(c) 4

(d) 5

79. If ||x + 3| + 1| = 1, then values of x is/are

(a) 3

(b) −1

(c) − 3

(d) 2

80. Solve for x :

$|x|^2 - 3|x| + 2 = 0$

(a) $x \in \{-2, -1, -1, 2\}$

(b) $x \in \{-2, -1, 1, -2\}$

(c) $x \in \{2, -1, 1, 2\}$

(d) $x \in \{-2, -1, 1, 2\}$

81. If $||x - 3| - 2| = 1$, then value of x is/are

(a) 1 or 3

(b) 3 or 5

(c) -3 or 5

(d) 0 or 2 or 4 or 6

82. If $x^2 - 4|x| + 3 = 0$, then values of x is/are

(a) ± 2 or ± 4

(b) 0 or ± 1

(c) ± 1 or ± 3

(d) ± 2 or 0

83. Solve for x :

$\left[x - \frac{1}{2}\right] = 2$, where [.] represents GIF

(a) $x \in [2.5, 3.5)$

(b) $x \in (2.5, 3.5)$

(c) $x \in [2.5, 3.5]$

(d) $x \in (2.5, 3.5]$

84. Solve for x

[2x − 3] = 1.5

(a) x ∈ R

(b) x ∈ Z

(c) x = 0

(d) No solution

85. Find the value of $\left[\frac{1}{2}\right] + \left[\frac{1}{2} + \frac{1}{1000}\right] + \ldots\ldots \left[\frac{1}{2} + \frac{2946}{1000}\right]$ where [.] denotes greatest integer function?

86. If the solution of the equation |2x − 1| = 3[x] + 2 {x} is p, where [.] denotes greatest integer and [.] denotes fractional part function then 8P is _____.

87. Let f : [π, 3π/2] → R be a function given by f(x) = [sin x] + [1 + sin x] + [2 + sin x] Then, the range of f(x) is

(a) {0, 3}

(b) {1}

(c) {0, 2}

(d) {3}

88. If [x] and {x} represent integral and fractional prats of x, then the expression [x] + $\sum_{r=1}^{2000} \frac{[x+r]}{2000}$ is equal to

(a) $\frac{2001}{2}x$

(b) $x + 2001$

(c) x

(d) $[x] + \frac{2001}{2}$

89. If the function f : R → R be such that f(x) = x − [x], where [x] denotes the greatest integer less than or equal to x, then $f^{-1}(x)$, is

(a) $\frac{1}{x-[x]}$

(b) [x] − x

(c) Not defined

(d) None of these

90. If f : R → R and g : R → R are defined by f(x) = x − [x] and g(x) = x − [x] for x ∈ R, where [x] is the greatest integer not exceeding x, then for every x ∈ R, f (g(x)) is equal to

(a) x

(b) 0

(c) f(x)

(d) g(x)

Answer Key

1. B	**2.** A	**3.** D	**4.** C	**5.** A	**6.** A
7. C	**8.** C	**9.** B	**10.** A	**11.** C	**12.** B
13. A	**14.** B	**15.** C	**16.** A	**17.** C	**18.** B
19. C	**20.** B	**21.** B	**22.** A	**23.** A	**24.** D
25. 32	**26.** A	**27.** B	**28.** C	**29.** A	**30.** D
31. B	**32.** C	**33.** B	**34.** B	**35.** A	**36.** A
37. B	**38.** C	**39.** B	**40.** C	**41.** A	**42.** C
43. B	**44.** B	**45.** A	**46.** B	**47.** D	**48.** B
49. D	**50.** B	**51.** B	**52.** D	**53.** C	**54.** A
55. D	**56.** A	**57.** A	**58.** B	**59.** D	**60.** A
61. A	**62.** D	**63.** A	**64.** C	**65.** A	**66.** B
67. 3	**68.** 4	**69.** 2	**70.** A	**71.** D	**72.** C
73. B	**74.** A	**75.** B	**76.** A	**77.** D	**78.** B
79. C	**80.** D	**81.** D	**82.** C	**83.** A	**84.** D
85. 4341	**86.** 2	**87.** A	**88.** C	**89.** C	**90.** B

Solutions

1. (B) Given, $a - \frac{1}{a} = 6$

Squaring on both sides, we get

$a^2 + \frac{1}{a^2} - 2 = 36$

$a^2 + \frac{1}{a^2} = 38$

2. (A) $a - \frac{1}{a} = 6$

Squaring both sides

$a^2 + \frac{1}{a^2} - 2 = 36$

$a^2 + \frac{1}{a^2} = 38$

Again, squaring on both sides

$\left(a^2 + \frac{1}{a^2}\right)^2 = (38)^2$

$a^4 + \frac{1}{a^4} + 2 = 1444$

$a^4 + \frac{1}{a^4} = 1442$

3. (D) Given, $a + \frac{1}{a} = \sqrt{2}$

Squaring on both sides

$a^2 + \frac{1}{a^2} + 2 = 2$

Again, squaring on both sides

$a^4 + \frac{1}{a^4} + 2 = 0$

$a^4 + \frac{1}{a^4} = -2$

Which is not possible

So, no solution

4. (C) Given, $x^2 + \frac{1}{x^2} = 83$

Then, $\left(x - \frac{1}{x}\right)^2 + 2 = 83$

$\left(x - \frac{1}{x}\right)^2 = 81$

$x - \frac{1}{x} = \sqrt{81}$

$x - \frac{1}{x} = \pm 9$

5. (A) $Given,\ x^2 + \frac{1}{x^2} = 34$

$Then,$

$$\left(x+\frac{1}{x}\right)^2-2=34$$

$$\left(x+\frac{1}{x}\right)^2=36$$

$$\left(x+\frac{1}{x}\right)=\pm 6$$

6. (A) Given, $6\left(x^2+\frac{1}{x^2}\right)-35\left(x+\frac{1}{x}\right)+62=0.$

$$x+\frac{1}{x}=y$$

$$\Rightarrow \left(x+\frac{1}{x}\right)^2=x^2+\frac{1}{x^2}+2=y^2$$

$$\Rightarrow x^2+\frac{1}{x^2}=y^2-2$$

$$\therefore\ 6\left(y^2-2\right)-35y+62=0$$

$$\Rightarrow 6y^2-35y+50=0$$

$$\Rightarrow 6y^2-20y-15y+50=0$$

$$\Rightarrow (2y-5)(3y-10)=0$$

$$\Rightarrow y=\tfrac{5}{2},\tfrac{10}{3}$$

$$\Rightarrow x+\tfrac{1}{x}=\tfrac{5}{2}\ or\ x+\tfrac{1}{x}=\tfrac{10}{3}$$

$$\Rightarrow 2x^2-5x+2=0\ or\ 3x^2-10x+3=0$$

$$\Rightarrow (x-2)(2x-1)=0\ \ or\ (x-3)(3x-1)=0$$

$$\Rightarrow x=2,\tfrac{1}{2},3,\tfrac{1}{3}$$

7. (C) $x^2+\frac{1}{x^2}-3x-\frac{3}{x}+4=0$

$$\Rightarrow \left(x+\frac{1}{x}\right)^2-2-3\left(x+\frac{1}{x}\right)+4=0$$

Substituting t = x + $\frac{1}{x}$, we get

$$t^2-3t+2=0\Rightarrow (t-1)(t-2)=0\Rightarrow t=1,2$$

$$\therefore x+\frac{1}{x}=1\ or\ x+\frac{1}{x}=2$$

$$\Rightarrow x^2-x+1=0\ or\ x^2-2x+1=0$$

$$\Rightarrow x=1$$

8. (C) Given, $2x^2-9x+14-\frac{9}{x}+\frac{2}{r^2}=0$

$$\Rightarrow 2\left(x^2+\frac{1}{x^2}\right)-9\left(x+\frac{1}{x}\right)+14=0$$

$$x+\frac{1}{x}=y\Rightarrow x^2+\frac{1}{x^2}=y^2-2$$

$$\therefore 2\left(y^2-2\right)-9y+14=0$$

$$\Rightarrow 2y^2-9y+10=0$$

$$\Rightarrow 2y^2-4y-5y+10=0$$

$\Rightarrow (2y-5)(y-2)=0$

$\Rightarrow y=\frac{5}{2},2$

$\therefore x+\frac{1}{x}=\frac{5}{2}\ or\ x+\frac{1}{x}=2$

$\Rightarrow 2x^2-5x+2=0\ or\ x^2-2x+1=0$

$\Rightarrow x=2,\frac{1}{2},1$

9. (B)

Given,

$3x^4-4x^3-14x^2-4x+3=0$

$\Rightarrow 3x^2-4x-14-\frac{4}{x}+\frac{3}{x^2}=0$

$\Rightarrow 3\left(x^2+\frac{1}{x^2}\right)-4\left(x+\frac{1}{x}\right)^x-14=0$

$x+\frac{1}{x}=y\Rightarrow x^2+\frac{1}{x^2}=y^2-2$

$\therefore 3\left(y^2-2\right)-4y-14=0$

$\Rightarrow 3y^2-4y-20=0$

$\Rightarrow 3y^2+6y-10y-20=0$

$\Rightarrow (3y-10)(y+2)=0$

$\Rightarrow y=\frac{10}{3},-2$

$\therefore\ x+\frac{1}{x}=\frac{10}{3}\ or\ x+\frac{1}{x}=-2$

$\Rightarrow 3x^2-10x+3=0\ or\ x^2+2x+1=0$

$\Rightarrow x=3,\ \frac{1}{3},\ -1$

10. (A) Given, $x^2+\frac{1}{x^2}-3\left(x-\frac{1}{x}\right)=0$

$x-\frac{1}{x}=y\Rightarrow x^2+\frac{1}{x^2}=y^2+2$

$\therefore\ y^2+2-3y=0$

$\Rightarrow (y-1)(y-2)=0$

$\Rightarrow y=1,\ 2$

$\therefore\ x-\frac{1}{x}=1\ or\ x-\frac{1}{x}=2$

$\Rightarrow\ x^2-x-1=0\ or\ x^2-2x-1=0$

$\Rightarrow x=\frac{1\pm\sqrt{5}}{2},\ \frac{2\pm\sqrt{8}}{2}$

$\therefore\ x=\frac{1\pm\sqrt{5}}{2},\ 1\pm\sqrt{2}$

11. (C) Given, $x^4-4x^3+x^2+4x+1=0$

$\Rightarrow x^2-4x+1+\frac{4}{x}+\frac{1}{x^2}=0$

$\Rightarrow x^2+\frac{1}{x^2}-4\left(x-\frac{1}{x}\right)+1=0$

$x-\frac{1}{x}=y\Rightarrow x^2+\frac{1}{x^2}=y^2+2$

$\therefore y^2+2-4y+1=0$

$\Rightarrow y^2-4y+3=0$

$\Rightarrow (y-1)(y-3)=0$

$\Rightarrow y=1,3$

$\therefore x-\frac{1}{x}=1\ or\ x-\frac{1}{x}=3$

$\Rightarrow x^2-x-1=0\ or\ x^2-3x-1=0$

$\Rightarrow x=\frac{1\pm\sqrt{5}}{2},\ \frac{3\pm\sqrt{13}}{2}$

12. (B) Given, $x^2 + a^2 + 9 - xa - 3a - 3x = 0$

$(x^2) + (a)^2 + (3)^2 - (a)\,(x) - (3)\,(a) - (3)\,(x) = 0$

$\frac{1}{2}\left((x-a)^2+(a-3)^2+(x-3)^2\right)=0$

Thus, $x = a = 3$

13. (A) Given, $4a^2+9b^2+16c^2=6ab+8ac+12bc$

$\Rightarrow 4a^2+9b^2+16c^2-6ab-8ac-12bc=0$

$\Rightarrow (2a)^2+(3b)^2+(4c)^2-(2a)(3b)-(2a)(4c)-(3b)(4c)=0$

$\Rightarrow 2(2a)^2 + 2(3b)^2 + 2(4c)^2 - 2(2a)(3b) - 2(2a)(4c) - 2(3b)(4c) = 0.$

$\Rightarrow (2a)^2 - 2(2a)(3b) + (3b)^2 + (3b)^2 - 2(3b)(4c) + (4c)^2 + (4c)^2 - 2(2a)(4c) + (2a)^2 = 0$

$\Rightarrow (2a - 3b)^2 + (3b - 4c)^2 + (4c - 2a)^2 = 0$

$\Rightarrow 2a - 3b = 3b - 4c = 4c - 2a = 0$

$\Rightarrow 2a = 3b = 4c$

14. (B) Given, $4x^2 + 9y^2 + z^2 = 6xy + 3yz + 2xz$

$(2x)^2 + (3y)^2 + z^2 - (2x)(3y) - (3y)z - (2x)z = 0$

Thus, 2x = 3y = z

15. (C) Given, $a^2 + \frac{b^2}{4} + 4c^2 - \frac{ab}{2} - bc - 2ac = 0$

$\Rightarrow 2\left(a^2\right) + 2\left(\frac{b^2}{4}\right) + 2\left(4c^2\right) - 2(a)\left(\frac{b}{2}\right) - 2\left(\frac{b}{2}\right)(2c) - 2(a)(2c) = 0$

$\Rightarrow a^2 - 2(a)\left(\frac{b}{2}\right) + \frac{b^2}{4} + \frac{b^2}{4} - 2\left(\frac{b}{2}\right)(2c) + 4c^2 + 4c^2 - 2(a)(2c) + a^2 = 0$

$\Rightarrow \left(a - \frac{b}{2}\right)^2 + \left(\frac{b}{2} - 2c\right)^2 + (2c - a)^2 = 0$

$\Rightarrow a - \frac{b}{2} = \frac{b}{2} - 2c = 2c - a = 0$

$\Rightarrow a = \frac{b}{2} = 2c$

16. (A) $a^2 + b^2 + c^2 = bc + ac + ac$ (Given)

$\therefore a^2 + b^2 + c^2 - ab - bc - ac = 0$

$\therefore a = b = c$

17. (C) $\left(a + \frac{1}{a}\right)^2 = 16$

$\therefore a^2 + \frac{1}{a^2} + 2 = 16$

$\therefore a^2 + \frac{1}{a^2} = 14$

$(a + b + c)^2 = 36$

18. (B) $a^2 + b^2 + c^2 + 2(ab + bc + ca) = 36$

$24 + 2(ab + bc + ca) = 36$

$\therefore ab + bc + ca = 6$

19. (C) $x^2 + 4y^2 + 16 - 2xy - 4x - 8y = 0$

$x^2 + (2y)^2 + (4)^2 - x(2y) - x(4) - (2y)(4) = 0$

$\therefore \quad x = 2y = 4$

20. (B) $(a + 2b + c)^2 = 64$

$\therefore a^2 + 4b^2 + c^2 + 2(2ab + 2bc + ac) = 64$

$\therefore \quad 50 + 2(2ab + 2bc + ac) = 64$

$\therefore \quad 2ab + 2bc + ac = 7$

21. (B) Sum of 3 perfect squares = 0

Means each term is zero

i.e., a – b = 0 & b – c = 0 & c – a = 0

∴ a = b & b = c & c = a

∴ a = b = c

22. (A) a > b

Then adding 5 on both sides

∴ a + 5 > b + 5

23. (A) 2a > 2b

24. (D) As multiplying or dividing by same negative real number on both sides of an inequality reverses the sign of the inequality, so option (d) is correct

25. (32) Given, $\log_{\sqrt{8}} b = \frac{10}{3}$

$\frac{\log_e b}{\log_e \sqrt{8}} = \frac{10}{3}$

Now, $\log_e \sqrt{8} = \log_e (2)^{\frac{3}{2}}$

$[\because \ \log_e m^n = n \log_e m]$

$\Rightarrow \ \log_e b = \frac{10}{3} \times \log_e (2)^{\frac{3}{2}}$

$\Rightarrow \log_e b = \frac{10}{3} \times \frac{3}{2} \times \log_e 2$

$\Rightarrow \log_e b = 5 \log_e 2$

$\Rightarrow \log_e b = \log_e 2^5$

Taking antilog on both sides

$b = 2^5 = 32$

26. (A) Given,

$3\log_{10} 1.5 + \log_{10} 240 - 2\log_{10} 9$

$\log_{10} (1.5)^3 + \log_{10} 240 - \log_{10} 9^2$

$\log_{10} \frac{(1.5)^3 (240)}{9^2}$

$\log_{10} \left(\frac{15^3 \times 240}{10^3 \cdot 9^2} \right) = \log_{10} 10 = 1$

27. (B) Given,

$3 \log 4 - 2 \log 6 + \log (18)^{3/2}$

$\log 4^3 - \log 6^2 + \log 18^{\frac{3}{2}}$

$\log \frac{4^3 \times 18^{\frac{3}{2}}}{6^2}$

$\log 96\sqrt{2}$

28. (C) log 3 + log(x + 1) = log (2x + 7)

log (3(x + 1)) = log (2x + 7)

3x + 3 = 2x + 7

x = 4

29. (A) $\log 2 + \log(x+2) - \log(3x-5) = \log 3$

$\log\left(\frac{x+2}{3x-5}\right) = \log\left(\frac{3}{2}\right)$

$\frac{x+2}{3x-5} = \frac{3}{2}$

$2x + 4 = 9x - 15$

$19 = 7x$

$x = \frac{19}{7}$

30. (D) $\log_4 \log_3 \log_2 x = 0$

$\log_3 \log_2 x = 4^0$

$\log_2 x = 3^1$

$x = 2^3$

x = 8

31. (B) $\log_{10} 1000 - \log_{10} 100 = \log_{10} \frac{1000}{100}$

$= \log_{10} 10$

= 1

32. (C) Using $\log_a mn = \log_a m + \log_a n$

We can say that,

$\log_{10} 5 + \log_{10} 20 = \log_{10}(5 \times 20)$

$= \log_{10} 100$ = 2

33. (B) Writing 1 as $\log_2 2$ to simplify the denominator, we get

$$\frac{\log_2 7}{\log_2 2 + \log_2 3} = \frac{\log_2 7}{\log_2 (2 \times 3)} = \frac{\log_2 7}{\log_2 6}$$

34. (B) $2\log_4 3 + \log_4 \frac{x^2}{9} = 1$

$\Rightarrow \log_4 3^2 + \log_4 x^2 - \log_4 9 = 1$

$\Rightarrow \log_4 9 + \log_4 x^2 - \log_4 9 = 1$

$\Rightarrow \log_4 x^2 = 1$

$\Rightarrow x^2 = 4$

$\Rightarrow |x| = 2$

35. (A) 2 + log x = log 500

$\Rightarrow \log 10^2 + \log x = \log 500$

$\Rightarrow \log 100 + \log x = \log 500$

$\Rightarrow \log 100x = \log 500$

$\Rightarrow 100x = 500$

$\Rightarrow x = 5$

36. (A) Let x be the required logarithm, then by definition $(2\sqrt{2})^x = 32\sqrt[5]{4}$

$\Rightarrow \left(2.2^{1/2}\right)^x = 2^5.2^{2/5};$

$\therefore 2^{\frac{3x}{2}} = 2^{5+\frac{2}{5}}$

Here, by equating the indices, $\frac{3}{2}x = \frac{27}{5}$

$\therefore x = \frac{18}{5} = 3.6.$

37. (B) $\log_3 4 \cdot \log_4 5 \cdot \log_5 6 \cdot \log_6 7 \cdot \log_7 8 \cdot \log_8 9$

$= \dfrac{\log 4}{\log 3} \cdot \dfrac{\log 5}{\log 4} \cdot \dfrac{\log 6}{\log 5} \cdot \dfrac{\log 7}{\log 6} \cdot \dfrac{\log 8}{\log 7} \cdot \dfrac{\log 9}{\log 8} = \dfrac{\log 9}{\log 3}$

$= \log_3 9 = \log_3 3^2 = 2$

38. (C) $\log_7 \log_7 \sqrt{7\sqrt{7\sqrt{7}}} = \log_7 \log_7 7^{7/8} = \log_7 \left(\dfrac{7}{8}\right)$

$= \log_7 7 - \log_7 8 = 1 - \log_7 2^3 = 1 - 3\log_7 2.$

39. (B) $a^2 + 4b^2 = 12ab$

$\Rightarrow a^2 + 4b^2 + 4ab = 16ab \Rightarrow (a + 2b)^2 = 16ab$

$\Rightarrow 2\log(a + 2b) = \log 16 + \log a + \log b$

$\therefore \log(a + 2b) = \frac{1}{2}[\log a + \log b + 4\log 2]$

40. (C)

$A = \log_2 \log_2 \log_4 256 + 2\log_{21/2} 2$

$= \log_2 \log_2 \log_4 4^4 + 2 \times \frac{1}{\left(\frac{1}{2}\right)} \log_2 2$

$= \log_2 \log_2 4 + 4 = \log_2 \log_2 2^2 + 4$

$= \log_2 2 + 4 = 1 + 4 = 5$

41. (A) $\dfrac{\log x}{b - c} = \dfrac{\log y}{c - a} = \dfrac{\log z}{a - b} = k(\text{say})$

$\Rightarrow \log x = k(b - c), \log y = k(c - a), \log z = k(a - b)$

$\Rightarrow x = e^{k(b-c)}, y = e^{k(c-a)}, z = e^{k(a-b)}$

$\therefore xyz = e^{k(b-c)+k(c-a)+k(a-b)} = e^0 = 1$

$x^a y^b z^c = e^{k(b-c)a+k(c-a)b+k(a-b)c} = e^0 = 1 = xyz$

$x^{b+c} y^{c+a} z^{a+b} = e^{k\left(b^2-c^2\right)+k\left(c^2-a^2\right)+k\left(a^2-b^2\right)} = e^0 = 1.$

42. (C) $\log_7 \log_5 \left(\sqrt{x^2 + 5 + x}\right) = 0 = \log_7 1$

$\Rightarrow \left(x^2 + 5 + x\right)^{1/2} = 5$

$\Rightarrow \left(x^2 + x + 5\right) = 25 \Rightarrow x^2 + x - 20 = 0$

$\Rightarrow (x - 4)(x + 5) = 0 \Rightarrow x = 4, -5$

$\Rightarrow x = 4$

43. (B)

$\Rightarrow x^2 - 2x - 3 = 0$

$\Rightarrow (x - 3)(x + 1) = 0$

$\Rightarrow x = -1,\ 3 (x = -1 \text{ reject } \because\ x > 0)$

Number of values of x is one.

44. (B) $\log_{10}\left(x^2 - 12x + 36\right) = 2$

$\Rightarrow x^2 - 12x + 36 = 100$

$\Rightarrow x^2 - 12x - 64 = 0$

$\Rightarrow (x - 16)(x + 4) = 0$

$\Rightarrow x = 16, -4$

45. (A) $\log_3\left(\log_9 x+\frac{1}{2}+9^x\right)=2x$

$\Rightarrow \log_9 x+\frac{1}{2}+9^x=3^{2x}$

$\Rightarrow \log_9 x+\frac{1}{2}+9^x=9^x$

$\Rightarrow \log_9 x=-\frac{1}{2}$

$\Rightarrow x=9^{-\frac{1}{2}}$

$\Rightarrow x=\frac{1}{\sqrt{9}}$

$\Rightarrow x=\frac{1}{3}$

46. (B) $x^2 > 4$

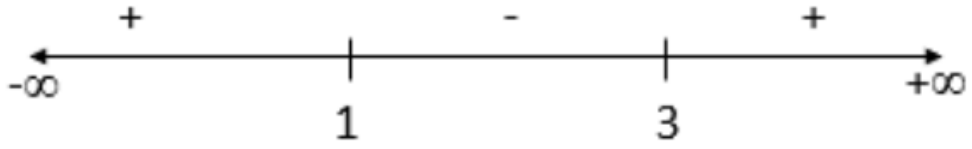
$\Rightarrow x^2-4>0$

$\Rightarrow (x+2)(x-2)>0$

$\Rightarrow x<-2 \, or \, x>2$

47. (D)

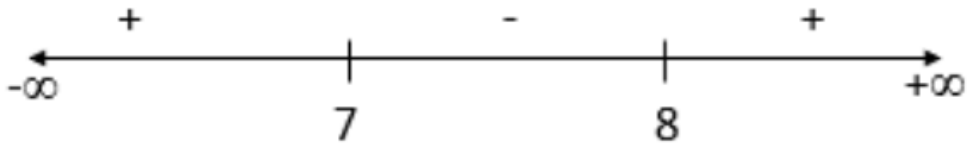
Given, (x − 1) (x − 3) > 0



$x \in (-\infty, 1) \cup (3, \infty)$

48. (B)

(x − 8) (x − 7) < 0

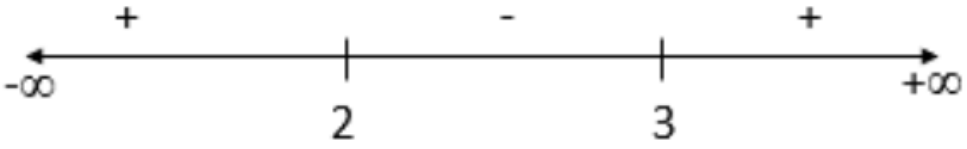


x ∈ (7, 8)

49. (D)

$x^2 - 5x + 6 \geq 0$

(x - 2) (x - 3) ≥ 0



$x \in (-\infty, 2] \cup [3, \infty)$

50. (B) $x^2 - 1 \leq 0$

$x^2 \leqslant 1$

$-1 \leqslant x \leqslant 1$

$x \in [-1, 1]$

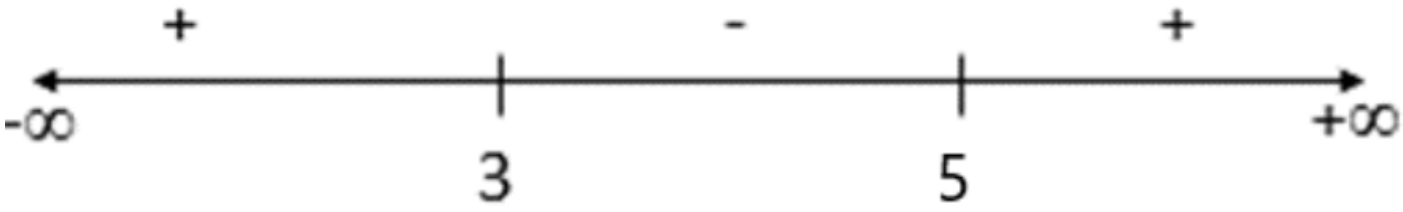
51. (B)

(x − 3) ≥ (x − 3) (x − 4)

$(x-3)[1-(x-4)] \geqslant 0$

$(x-3)(5-x) \geqslant 0$

$(x-3)(x-5) \leqslant 0$



$x \in [3, 5]$

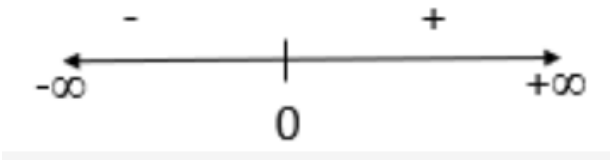
52. (D)

$$\frac{x+5}{x} \geqslant 1$$

$$\frac{x+5}{x} - 1 \geqslant 0$$

$$\frac{x+5-x}{x} \geqslant 0$$

$$\frac{5}{x} \geqslant 0$$



$x \in (0, \infty)$

53. (C)

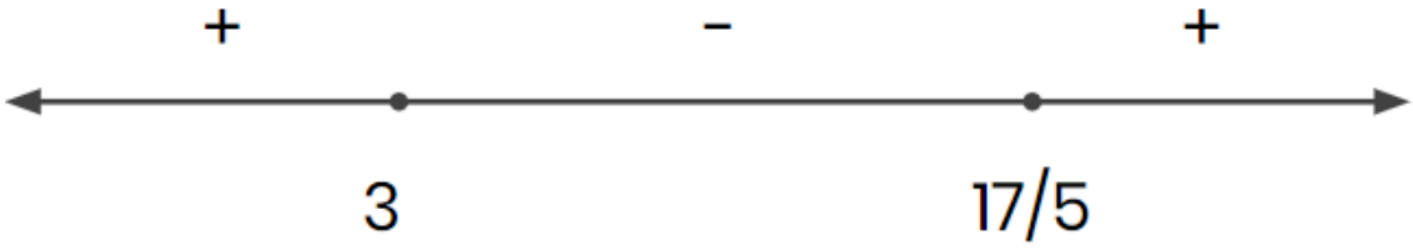
$$\frac{2}{x-3} \leqslant 5$$

$$\frac{2}{x-3} - 5 \leqslant 0$$

$$\frac{2-5x+15}{x-3} \leqslant 0$$

$$\frac{17-5x}{x-3} \leqslant 0$$

$$\frac{5x-17}{x-3} \geqslant 0$$



$x \in (-\infty, 3) \cup \left[\frac{17}{5}, \infty\right)$

54. (A) $x^2 + 4x + 4 \leq 0$

$(x + 2)^2 \leq 0$

It can't be negative

Thus, zero at x = − 2

55. (D) $x^2 + 4x + 4 > 0$

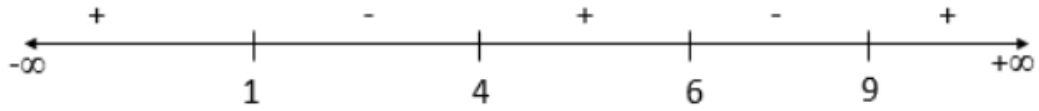
$(x + 2)^2 > 0$

It's always positive except value where x + 2 = 0

∴ R − {-2}

56. (A)

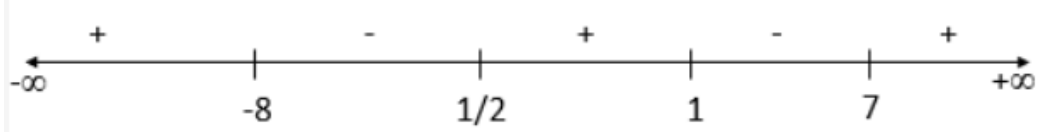
$(x-1)(x-4)(x-6)(x-9) \leqslant 0$



$x \in [1,4] \cup [6,9]$

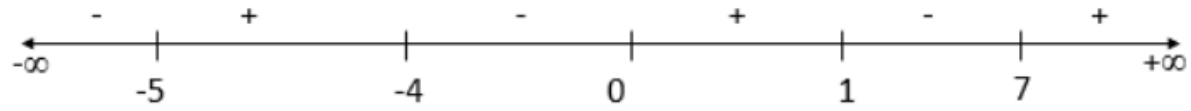
57. (A)

(2x – 1) (x – 1) (x + 8) (x – 7) ≤ 0



$x \in \left[-8, \frac{1}{2}\right] \cup [1,7]$

58. (B)



$x \in (-\infty,-5] \cup [-4,0) \cup [1,7)$

59. (D)

$\frac{x^2-5}{x-1} \geqslant -1$

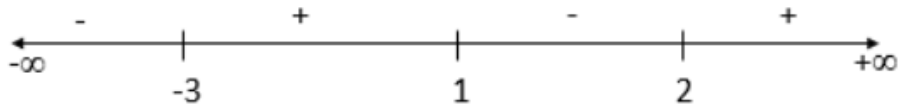
$\frac{x^2-5}{x-1} + 1 \geqslant 0$

$\frac{x^2-5+x-1}{x-1} \geqslant 0$

$\frac{x^2+x-6}{x-1} \geqslant 0$

$\frac{x^2+3x-2x-6}{x-1} \geqslant 0$

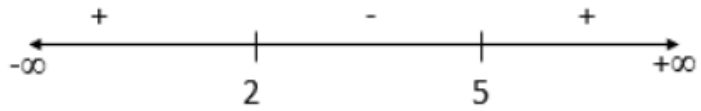
$\frac{(x+3)(x-2)}{(x-1)} \geqslant 0$



$x \in [-3,1) \cup [2,\infty)$

60. (A)

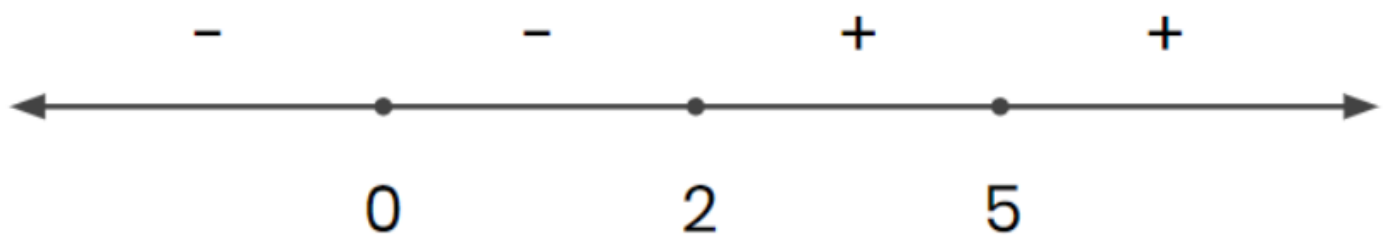
x^2 (x – 5) (x – 2) ≥ 0



$x \in (-\infty,2] \cup [5,\infty)$

61. (A)

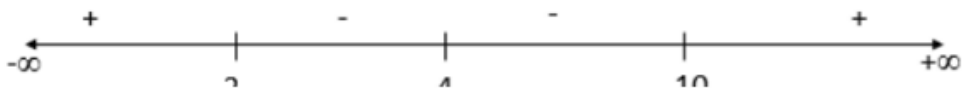
$\frac{x^2(x-5)^4}{(x-2)} < 0$



$x \in (-\infty,2) - \{0\}$

62. (D)

$(x-2)^3(x-4)^2(x-10)^{17} \geqslant 0$



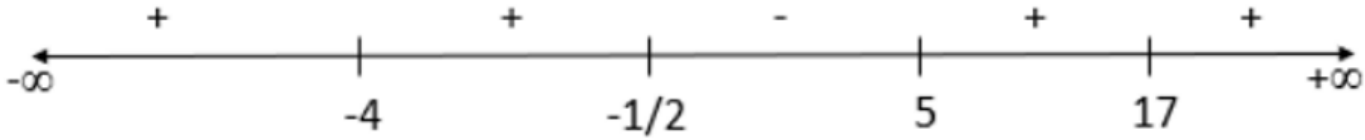
$x \in (-\infty, 2] \cup [10, \infty) \cup \{4\}$

63. (A) $(x-2)^4 (x-3)^2 (2x+1)^{24} \leq 0$

All even power never affects the inequation. Thus, Its always positive and zero at

$x = 2, 3, \frac{-1}{2}$

64. (C)



$\therefore x \in \left(-\infty, \frac{-1}{2}\right] \cup (5, 17) \cup (17, \infty)$

65. (A)

$\left(x^2+1\right)(x-2)^2(x-3) < 0$



$(-\infty, 2) \cup (2, 3)$

66. (B) $\left|x^2+1\right| = x^2+1$

True, as the equation holds for all real values of x.

67. (3) $\frac{x+2}{x-4} \leqslant 0 \Rightarrow x \in [-2, 4)$

$x^2 - ax - 4 \leqslant 0$

$\Rightarrow x \in \left[\frac{a-\sqrt{a^2+16}}{2}, \frac{a+\sqrt{a^2+16}}{2}\right]$

$\Rightarrow \frac{a+\sqrt{a^2+16}}{2} < 4 \Rightarrow a^2+16 < (a-8)^2$

$\Rightarrow a^2+16 < a^2-16a+64 \Rightarrow a < 3$

and $\frac{a-\sqrt{a^2+16}}{2} \geqslant -2 \Rightarrow a - \sqrt{a^2+16} \geqslant -4$

$(a+4)^2 \geqslant \sqrt{a^2+16} \Rightarrow a \geqslant 0$

$\therefore$ The possible integral values of a are 0, 1, 2

68. (4) The modulus of a given function gives the magnitude of that function. Modulus function is defined as the real valued function.

The real function $f : R \to R$ defined by $f(x) = |x| = x,\ x > 0$, and $f(x) = |x| = -x$, if $x < 0$. $\forall\, x \in R$ is called the modulus function.

Given $\left|x^2-12x+32\right| + \left|x^2-9x+20\right| = 0$.

Every modulus function is a non-negative function and if two non-negative functions add up to get zero then individual function itself equal to zero simultaneously.

$x^2 - 12x + 32 = 0$ for $x = 4$ or 8

$x^2 - 9x + 20 = 0$ for $x = 4$ or 5

Both the equations are zero at $x = 4$

So, $x = 4$ is the only solution for this equation.

69. (2) $|x| \geqslant 0$ for every $x \in R$

Let $f(x) = |x - 4| + 2$

As we know that $|x| \geqslant 0$ for every $x \in R$

$\therefore\ |x - 4| \geqslant 0$

The minimum value of function is attained when $|x - 4| = 0$

Hence, minimum value of $f(x) = 0 + 2 = 2$

70. (A) |x − 3| = 5

$\Rightarrow x - 3 = \pm 5$

$\Rightarrow x = 3 \pm 5$

$\Rightarrow x = 8, -2$

71. (D) |2x − 3| = − 5

|x| ≥ 0 ∀ x ∈ R

No Solution

72. (C) $|x^2 + 1| = 5$

$\Rightarrow x^2 + 1 = 5$

$\Rightarrow x^2 = 4$

$\Rightarrow x = \pm 2$

73. (B) x − 2 = 3 or x − 2 = − 3

∴ x = 5 or x = − 1

74. (A) x + 5 = 6 or x + 5 - − 6

x = 1 or x = − 11

75. (B) x^2 − 4x + 2 = 2 or x^2 − 4x + 2 = − 2

$\therefore\ x^2 - 4x = 0$ or $x^2 - 4x + 4 = 0$

$\therefore\ x(x - 4) = 0$ or $(x - 2)^2 = 0$

$\therefore\ x = 0$ or $x = 4$ or $x = 2$

76. (A) x^2 − 1 = 1 or x^2 − 1 = − 1

∴ x^2 = 2 or x^2 = 0

x = ± $\sqrt{2}$ or x = 0

77. (D) |x + 3| = − 3 is not possible because modulus of any number is never negative

78. (B) |x| = 3

∴ x = 3 or − 3

79. (C) |x + 3| + 1 = 1 or |x + 3| + 1 = − 1

∴ |x + 3| = 0 or |x + 3| = − 2 (Not possible)

x + 3 = 0

∴ x = − 3

80. (D) $|x|^2 - 3|x| + 2 = 0$

$|x| = x, x \geqslant 0$

$-x, x < 0$

$\text{For} |x| = x, x \geqslant 0$

$x^2 - 3x + 2 = 0$

$\Rightarrow (x-1)(x-2) = 0$

$\Rightarrow x = 1, 2$

$\therefore x \in \{1, 2\} \cap [0, \infty]$

$\Rightarrow x \in \{1, 2\}$

$\text{For} |x| = -x, x < 0$

$x^2 = 3x + 2 = 0$

$\Rightarrow (x+1)(x+2) = 0$

$\Rightarrow x = -1, -2$

$\therefore x \in \{-1, -2\} \cap [-\infty, 0]$

$\Rightarrow x \in \{-1, -2\}$

$\therefore x \in \{-2, -1, 1, 2\}$

81. (D) $|x-3| - 2 = 1 \; or \; |x-3| - 2 = -1$

$|x-3| = 3 \, or \; |x-3| = 1$

$\therefore \; x - 3 = 3 \; or \; -3 \; or \; x - 3 = 1 \; or \; -1$

∴ x = 6 or 0 or x = 4 or 2

82. (C) |x| = t

$t^2 - 4t + 3 = 0$

$(t-1)(t-3) = 0$

$\therefore t = 1 \, or \, 3$

$|x| = 1 \, or \, 3$

$x = \pm 1 \, or \, \pm 3$

83. (A) $\left[x - \frac{1}{2}\right] = 2$

$\Rightarrow x - \frac{1}{2} \in [2, 3)$

$\Rightarrow x \in [2.5, 3.5)$

84. (D) [2x − 3] = 1.5

[x] ∈ Z ⇒ No Solution

85. (4341)

$$\left[\frac{1}{2}\right]+\left[\frac{1}{2}+\frac{1}{1000}\right]+\ldots\ldots\left[\frac{1}{2}+\frac{499}{1000}\right]+\left[\frac{1}{2}+\frac{500}{1000}\right]+\ldots\ldots\left[\frac{1}{2}+\frac{1499}{1000}\right]+\left[\frac{1}{2}+\frac{1500}{1000}\right]+\ldots\ldots$$
$$+\left[\frac{1}{2}+\frac{2499}{1000}\right]+\left[\frac{1}{2}+\frac{2500}{1000}\right]+\ldots\ldots\left[\frac{1}{2}+\frac{2946}{1000}\right]$$
$$=0+1\quad 1000+2\quad 1000+3\quad 447=3000+1341=4341$$

86. (2) We are given that, |2x − 1| = 3[x] + 2{x}

Let, 2x − 1 ≤ 0 i.e. x ≤ $\frac{1}{2}$. The given equation yields.

1 − 2x = 3[x] + 2{x}

$$\Rightarrow \quad 1-2[x]-2\{x\}=3[x]+2\{x\} \quad \Rightarrow \quad 1-5[x]=4\{x\}\Rightarrow\{x\}=\frac{1-5[x]}{4}$$
$$\Rightarrow \quad 0\leqslant\frac{1-5[x]}{4}<1\Rightarrow 0\leqslant 1-5[x]<4\Rightarrow-\frac{3}{5}<[x]\leqslant\frac{1}{5}$$

Now, [x] = 0 as zero is the only integer lying between − $\frac{3}{5}$ < [x] ≤ $\frac{1}{5}$

⇒ {x} = $\frac{1}{4}\Rightarrow x=\frac{1}{4}$which is less than $\frac{1}{2}$, Hence $\frac{1}{4}$is one solution.

Now, let 2x − 1 > 0 i.e. x > $\frac{1}{2}$

$$\Rightarrow \quad 2x-1=3[x]+2\{x\} \quad \Rightarrow \quad 2[x]+2\{x\}-1=3[x]+2\{x\}$$
$$\Rightarrow \quad [x]=-1\Rightarrow -1\leqslant x<0$$ which is not a solution as x > $\frac{1}{2}$
$$\Rightarrow \quad x=\frac{1}{4}$$ is the only solution.

87. (A) For x ∈ (π , 3π /2) we have − 1 < sin x < 0

⇒ 0 < 1 + sin x < 1 and 1 < (2 + sin x) < 2

∴ [sin x] = -1, [1 + sin x] = 0 and [2 + sin x] = 1

⇒ f(x) = [sin x] + [1 + sin x] + [2 + sin x]

= 1 + 0 + 1 = 0

For x = π , we have

[sin x] = 0, [1 + sin x] = 1 and [2 + sin x] = 2

∴ f(x) =) + 1 + 2 = 3

For $x=\frac{3\pi}{2}$, we have

[sin x] = − 1, [1 + sin x] = 0 and [2 + sin x] = 1

∴ f(x) = − 1 + 0 + 1 = 0

Hence, range of f(x) = {0, 3}

88. (C) We have,

$$[\because [x+r]=[x]+r]$$

89. (C) We have, $f(x)=x-[x]$

$$\Rightarrow f(x)=\begin{cases}x-n,\ if\, n<x<n+1\\ n-n=0,\ if\, x=n\end{cases},\ where\, n\in Z$$

Thus, f(x) is a many one function

Consequently, $f^{-1}(x)$ is not defined

90. (B) Given, $f(x) = x - [x]$, $g(x) = [x]$ for $x \in R$.

$\therefore f(g(x)) = f([x])$

$= [x] - [x]$

$= 0$